

Paper Review

IMU2CLIP: MULTIMODAL CONTRASTIVE LEARNING FOR IMU MOTION SENSORS FROM EGOCENTRIC VIDEOS AND TEXT NARRATIONS

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Presented by: Hyeonseok Jung

Paper

- **Title:**

IMU2CLIP: Multimodal Contrastive Learning for IMU Motion Sensors from Egocentric Videos and Text

- **Authors:**

Meta Reality Labs (Seungwhan Moon, Andrea Madotto, Zhaojiang Lin, Alireza Dirafzoon, Aparajita Saraf, Amy Bearman, Babak Damavandi)

- **Conference:**

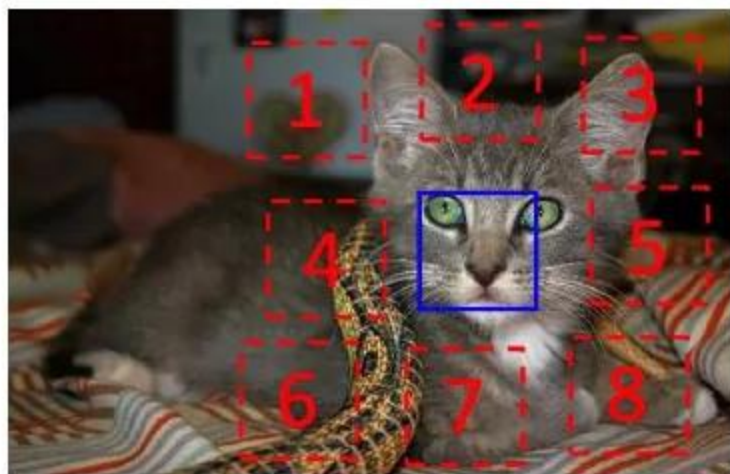
Findings of EMNLP 2023 (Findings of the Association for Computational Linguistics: EMNLP 2023)

- **Citations:**

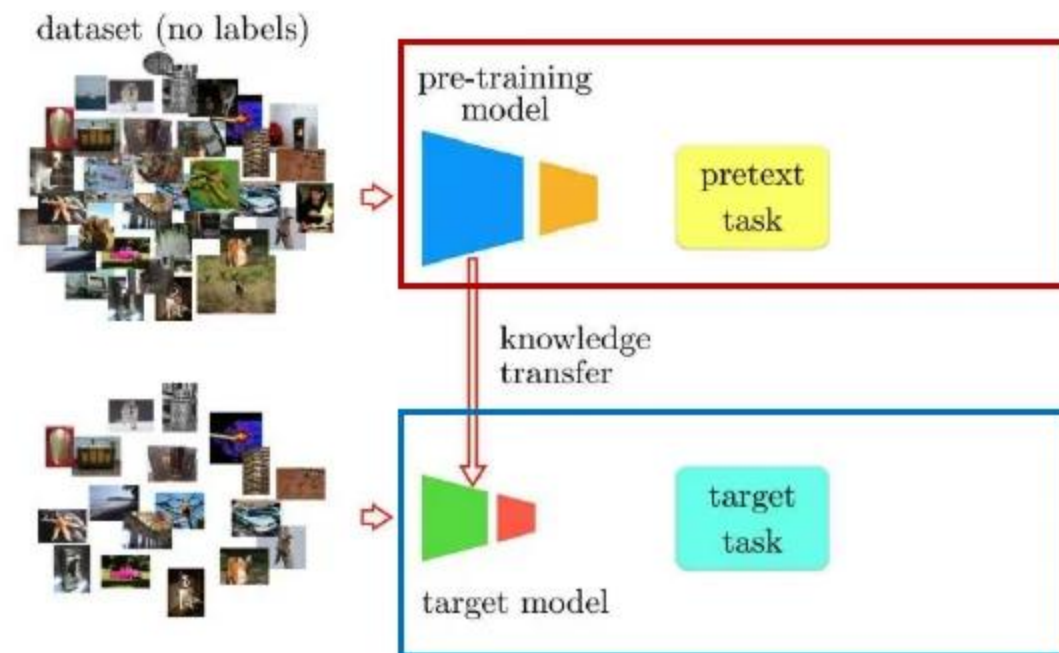
38 citations (as of July 2025, according to Google Scholar)

Background: Self-Supervised Learning

- Supervised learning performs well when there is a large amount of labeled data available.
- However, obtaining labeled data is often very expensive.
- Therefore, self-supervised learning **first solves a pretext task**, and then tackles the **target task**.



$$X = \left(\begin{array}{c} \text{cat face} \\ \text{cat ear} \end{array} \right); Y = 3$$



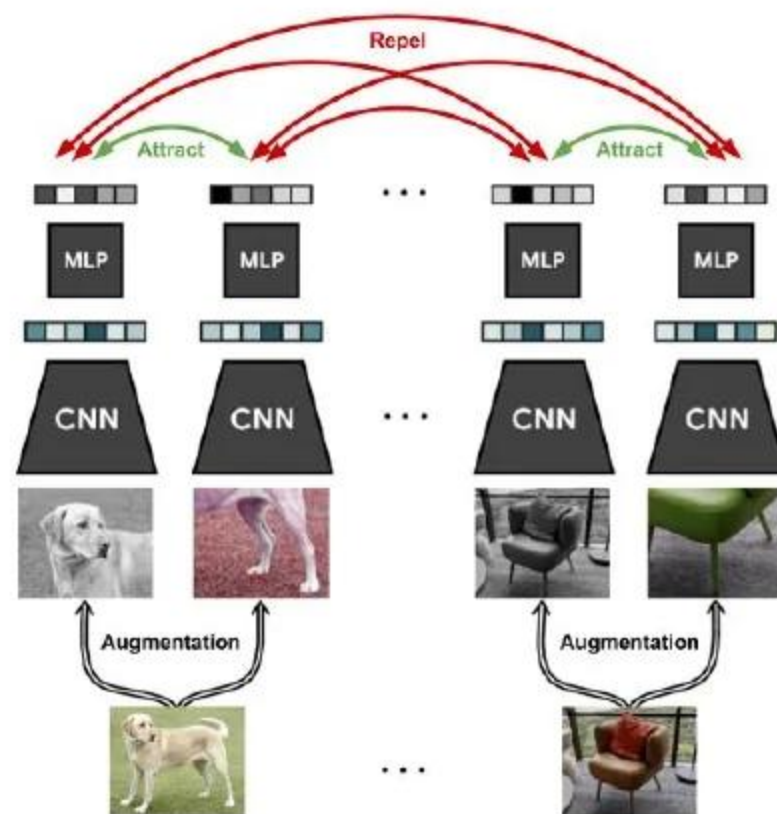
Background : Contrastive Learning

- The model learns to **make representations of images from the same source similar**, and **representations of images from different sources as dissimilar as possible**.



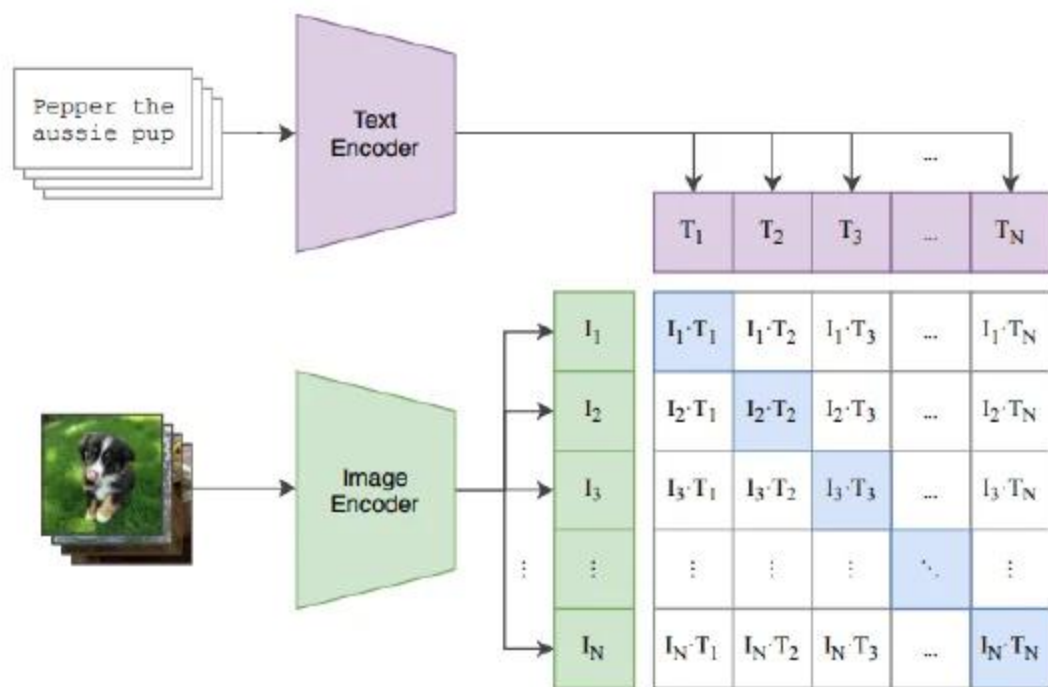
Model

?

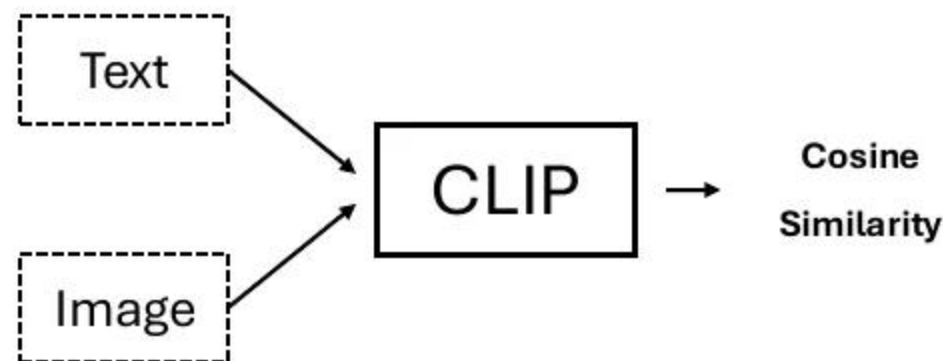


Background : CLIP

- CLIP(Contrastive Language-Image Pre-Training)
- This model **uses contrastive learning**, and applies it across different modalities—images and texts.
- By aligning image representations and text representations in a shared embedding space, it enables **joint learning between modalities**.



Cross-modal Understanding



Motivation & Intro

1. Opportunity:

The widespread adoption of **wearable IMU sensors** has **dramatically increased the value** and potential of motion data.

2. Problem:

Manual annotation-driven IMU models suffer from **high labeling costs** and poor generalization to new activities.

3. Inspiration:

Inspired by BERT/GPT and CLIP, **multimodal contrastive pre-training**.

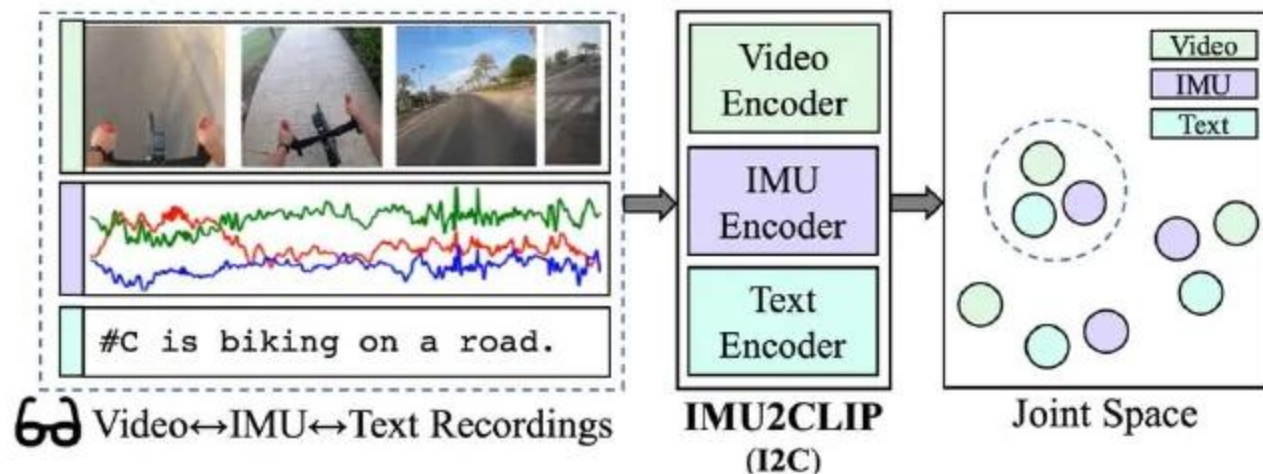


“What if we could pre-train IMU exactly like CLIP does for images and text—
and get rich, semantic embeddings without any labels?”

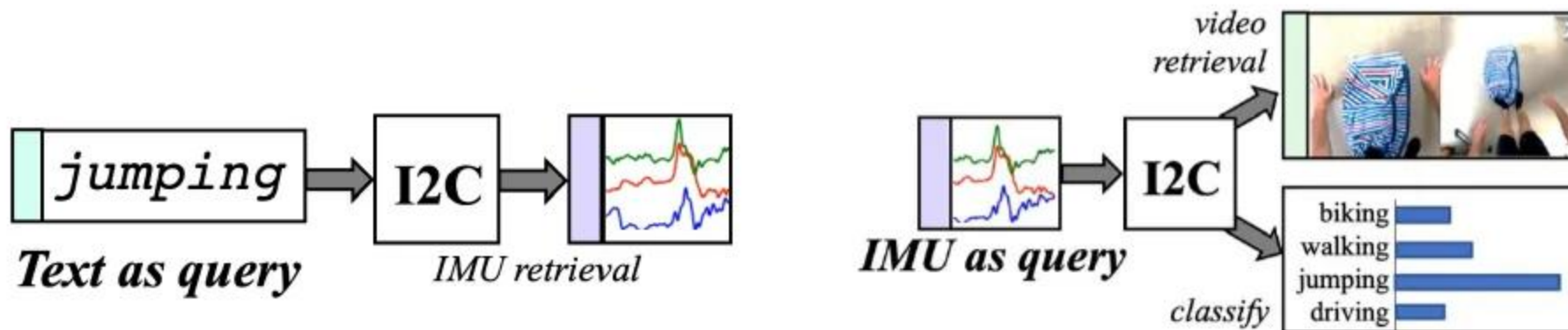
“ Let’s build an IMU-CLIP: multimodal IMU↔Video/Text contrastive pre-train! ”

Overall Framework

1. The model aligns the parallel “**video↔IMU↔text**” data in the joint space.



2. IMU and Videos, or as a classifier for **downstream applications**.

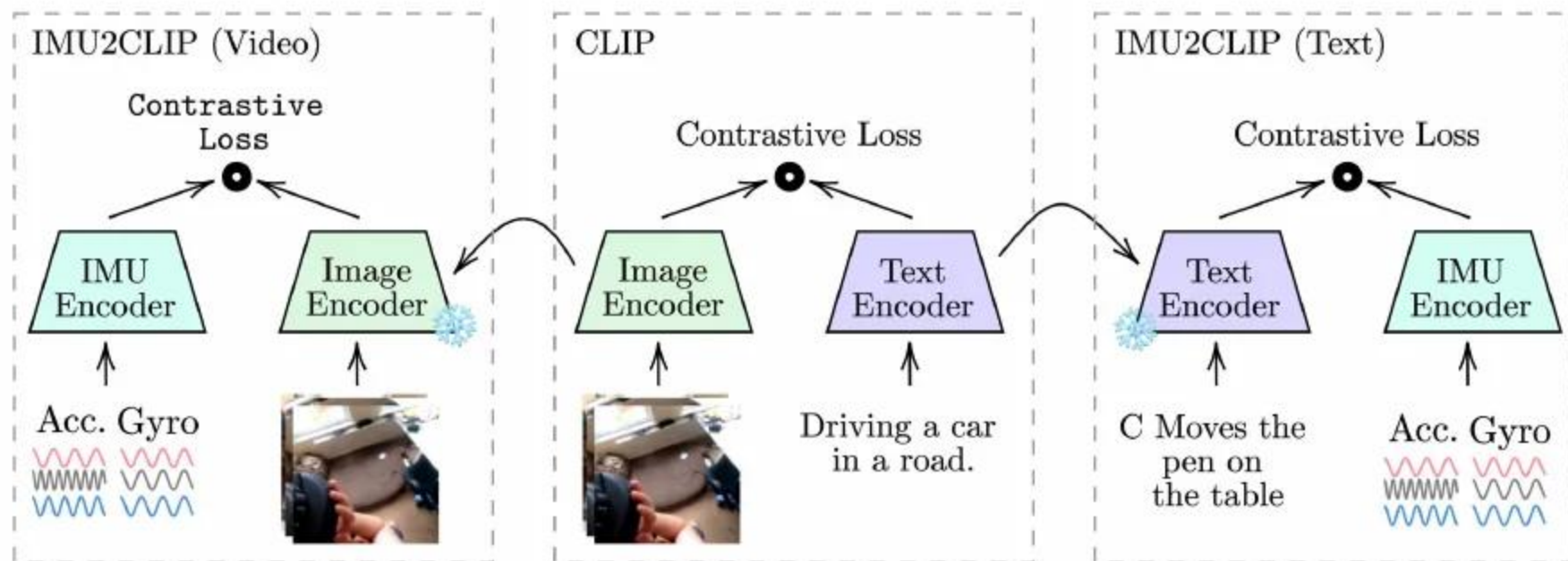


Methods

How IMU2CLIP Achieves Joint Space Alignment: **The Contrastive Learning**

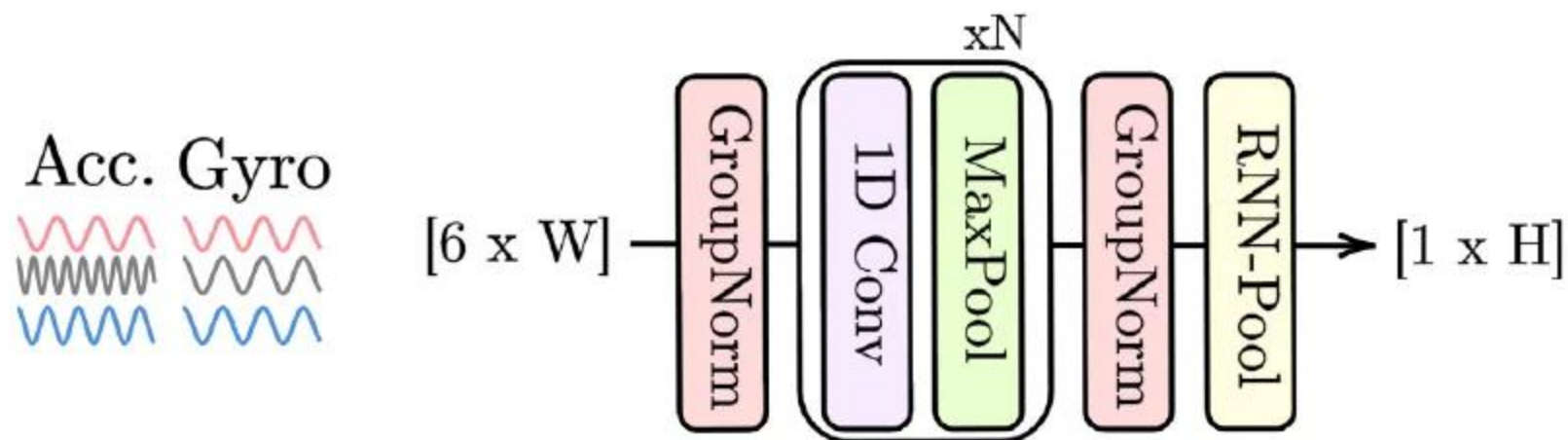
Paper achieve this comprehensive multi-modal integration by optimizing three distinct contrastive learning pairs

- IMU-Video Contrastive Loss (Left)
- Video-Text Contrastive Loss (Center)
- Text-IMU Contrastive Loss (Right)



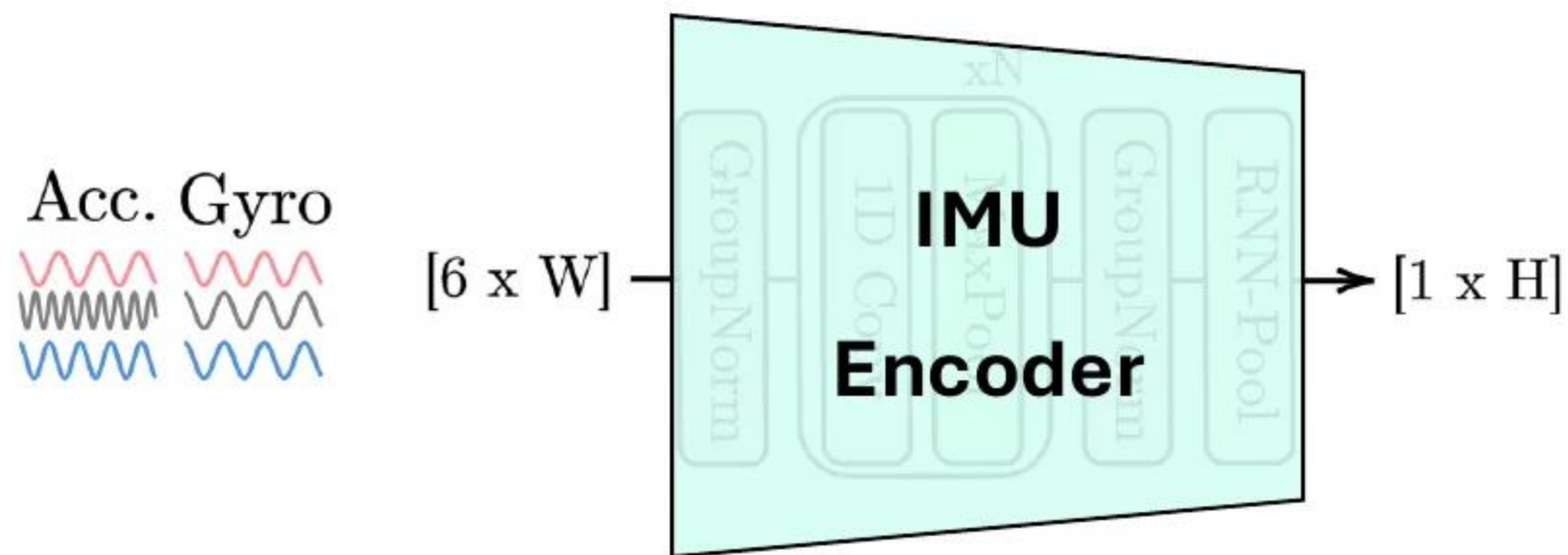
Methods

1. Encoder



Methods

1. Encoder



Methods

1. A batch of B ground-truth “**I**MU↔**V**ideo↔**T**ext” parallel windows:

$$\{(i_1, v_1, t_1), \dots, (i_B, i_B, i_B)\}$$

2. Calculate Similarity

$$\text{IMU, Image : } \text{sim}(i_i, v_j) = \langle i_i, v_j \rangle$$

$$\text{IMU, Text : } \text{sim}(i_i, t_j) = \langle i_i, t_j \rangle$$

3. Defining Probabilities for Alignment

$$P_{i2v}(v_j | i_i) = \frac{\exp(\sin(i_i, v_i))^{\frac{1}{\gamma}}}{\sum_{k=1}^B \exp(\sin(i_i, v_k))^{\frac{1}{\gamma}}}$$

Cross-Entropy

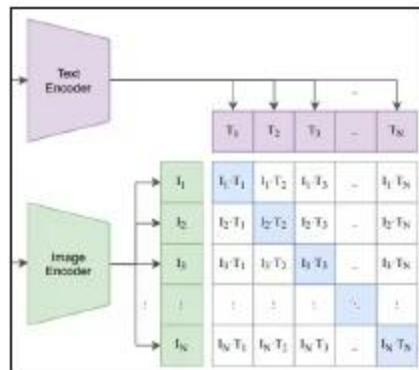
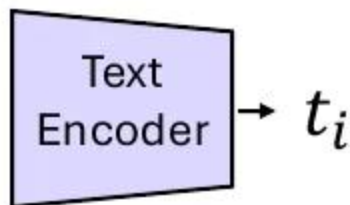
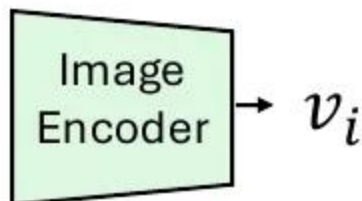
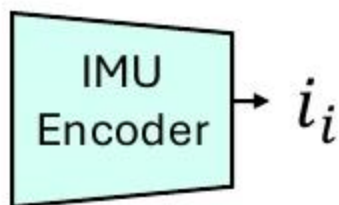
$$\sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

Where γ is a temperature parameter that controls the concentration of the distributions.

Acc. Gyro

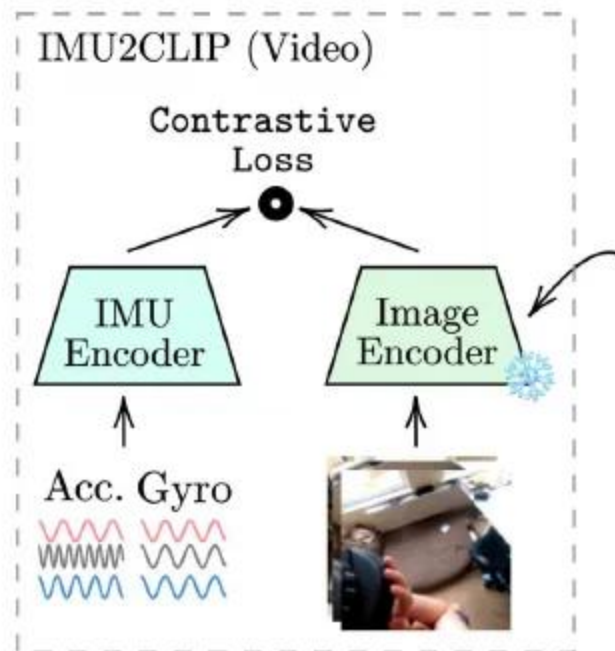


Driving a car
in a road.



Methods

4. Defining cross-entropy losses



$$IMU \text{ to Video: } \mathcal{L}_{i2v} = -\frac{1}{B} \sum_{i=1}^B \log\{P_{i2v}(v_j|i_i)\}$$

$$Video \text{ to IMU: } \mathcal{L}_{v2i} = -\frac{1}{B} \sum_{i=1}^B \log\{P_{v2i}(i_j|v_i)\}$$

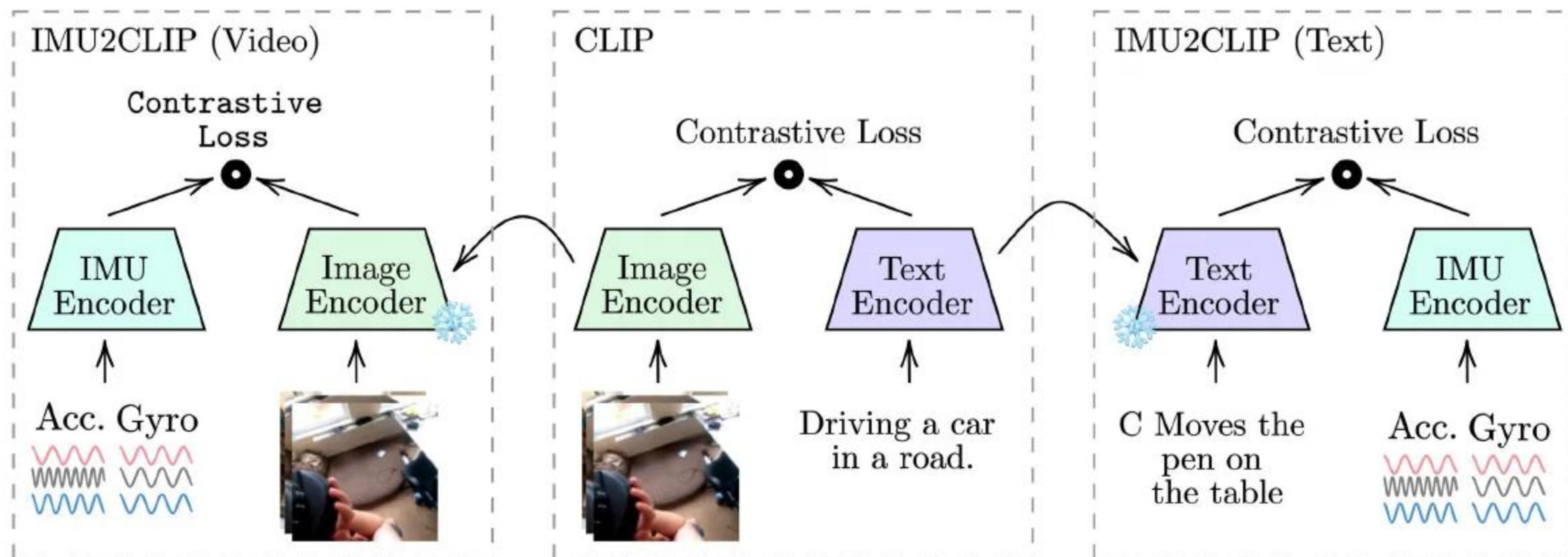
$$IMU \leftrightarrow Video: \mathcal{L}_{i \leftrightarrow v} = \frac{1}{2}(\mathcal{L}_{i2v} + \mathcal{L}_{v2i})$$



$$\mathcal{L}_{i \leftrightarrow v \leftrightarrow t} = \mathcal{L}_{i \leftrightarrow v} + \mathcal{L}_{i \leftrightarrow t}$$

Methods

$$\mathcal{L}_{i \leftrightarrow v \leftrightarrow t} = \mathcal{L}_{i \leftrightarrow v} + \mathcal{L}_{i \leftrightarrow t}$$



Experiments

0. Dataset

Ego4d



- Source: Bristol
- Scenario: Cleaning / laundry
- Topic: room, iron, shirt, fold, trouser



- Source: CMU
- Scenario: Construction/renovation jobs
- Topic: workshop, arrange, brick, walk



- Source: KAUST
- Scenario: Cleaning / laundry
- Topic: wash, dish, put, puzzle, dialogue

Aria

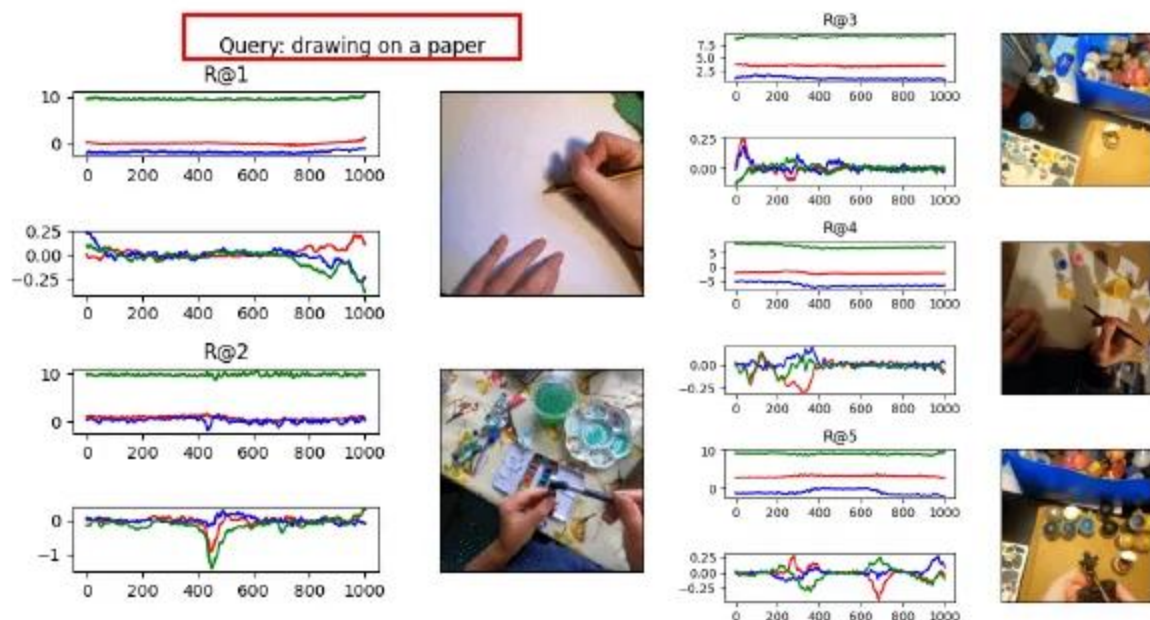


Ego4d	Tra.	Val.	Tst.
# of Media files	1444	161	688
Total Media Durations	540h	60h	265h
# of IMU↔Text/Video windows (5s)	528K	68K	266K
# of IMU→4 classes windows (5s)	1552	760	241

Aria	Tra.	Val.	Tst.
# of Media files	747	259	277
Total Media Durations	138h	43h	51h
# of IMU↔Video windows (1s)	496K	157K	184K
# of IMU→5 classes windows (1s)	25K	138K	162K

Experiments

1. Text $\rightarrow$ IMU : IMU-based media search with textual queries



Train Modalities			Text $\rightarrow$ IMU			
IMU	Video	Text	R@1($\uparrow$)	R@10($\uparrow$)	R@50($\uparrow$)	MRR($\uparrow$)
✓	✓		4.17	15.62	43.06	0.084
✓		✓	7.29	28.82	60.07	0.143
✓	✓	✓	5.90	22.92	56.60	0.139
			(Text $\rightarrow$ Video)			
(CLIP)	✓	✓	8.33	33.68	65.28	0.168

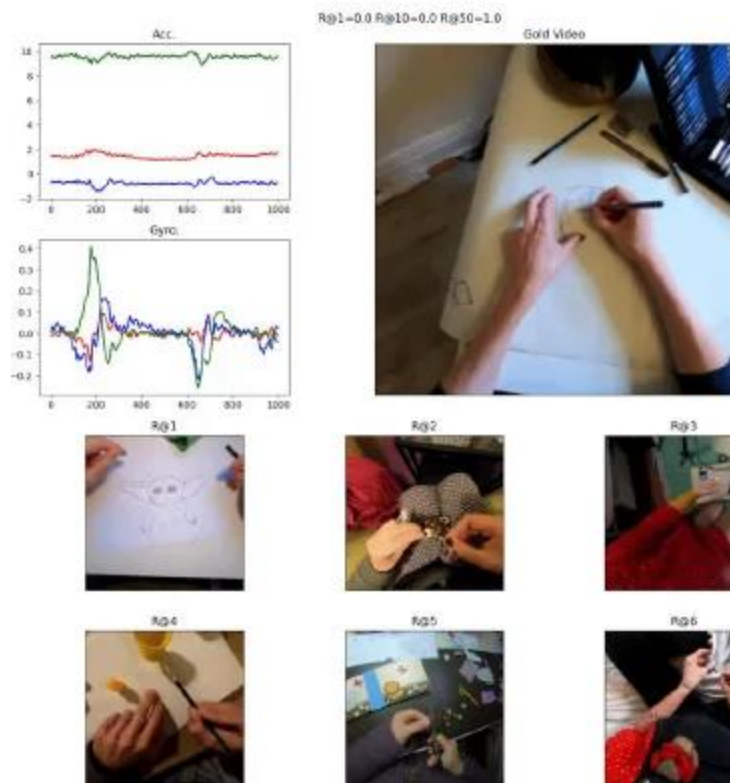
$\rightarrow$ Shows that the IMU encoder could serve as a power-efficient alternative for a video encoder in many applications.

R@K (Recall@K): Percentage of queries where the correct item is found within the **top K** retrieved results. (Higher is better $\uparrow$)

MRR (Mean Reciprocal Rank): Average of the reciprocal ranks of the correct item. The rank is 1 for the first correct item, 1/2 for the second, etc. (Higher is better $\uparrow$, Max 1)

Experiments

2. IMU $\rightarrow$ Video : search for videos, given IMU recordings



Result by Aria Dataset

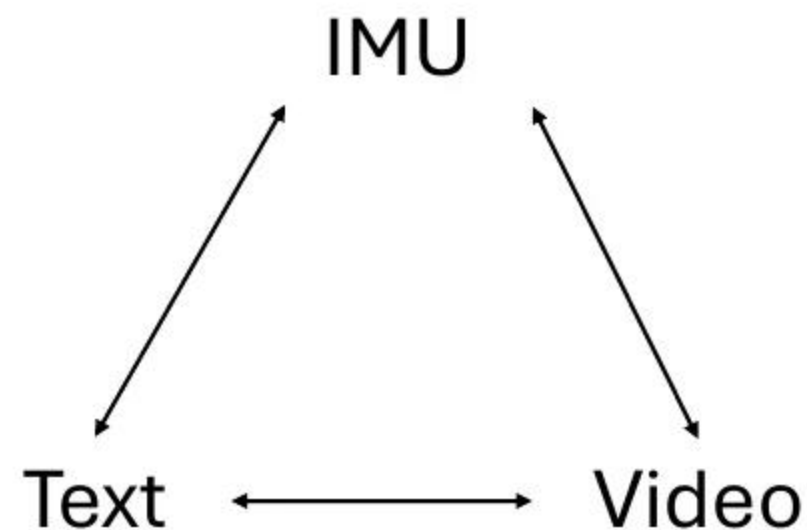
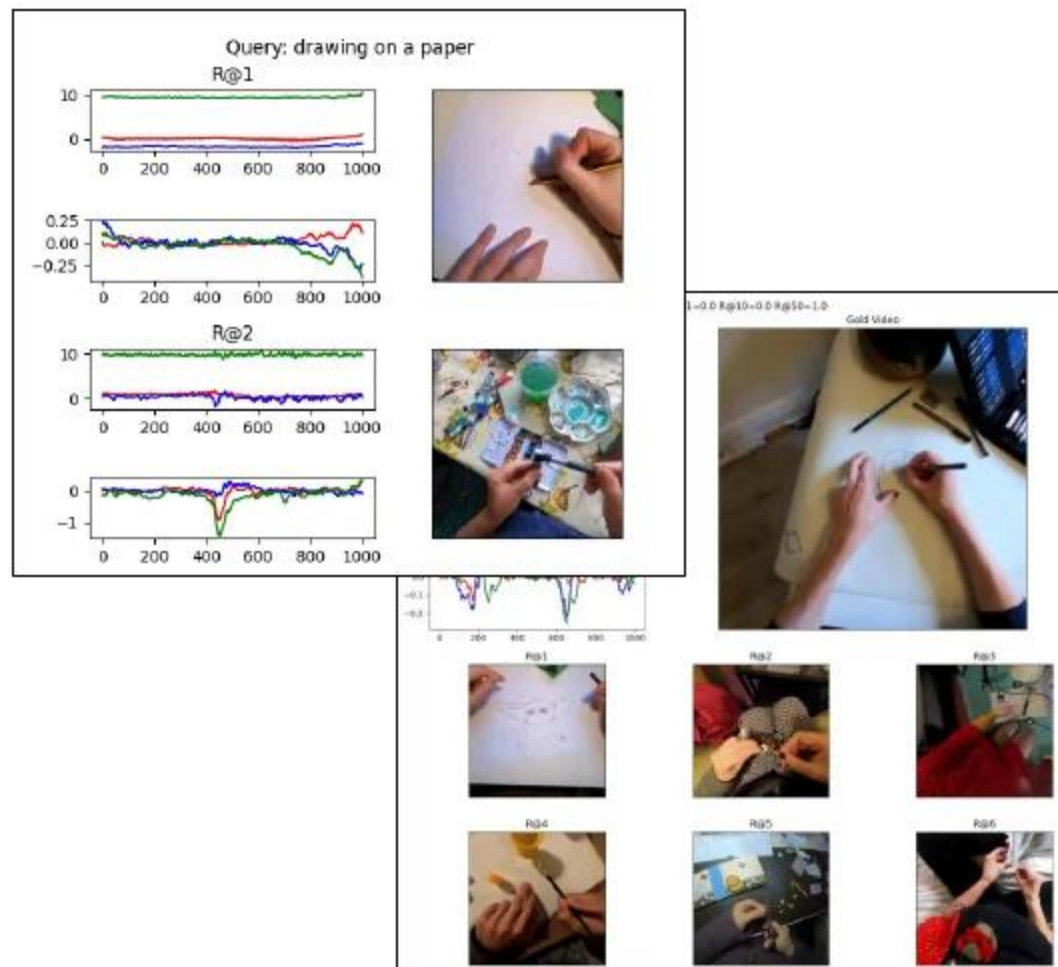
IMU $\rightarrow$ Video				Video $\rightarrow$ IMU			
$R@1(\uparrow)$	$R@10(\uparrow)$	$R@50(\uparrow)$	$MRR(\uparrow)$	$R@1(\uparrow)$	$R@10(\uparrow)$	$R@50(\uparrow)$	$MRR(\uparrow)$
8.48	38.83	77.67	0.182	8.44	44.19	78.57	0.190

$\rightarrow$ IMU2CLIP **effectively aligns IMU and Video** data even on the Aria dataset, which lacks text narrations.

R@K (Recall@K): Percentage of queries where the correct item is found within the **top K** retrieved results. (Higher is better $\uparrow$)

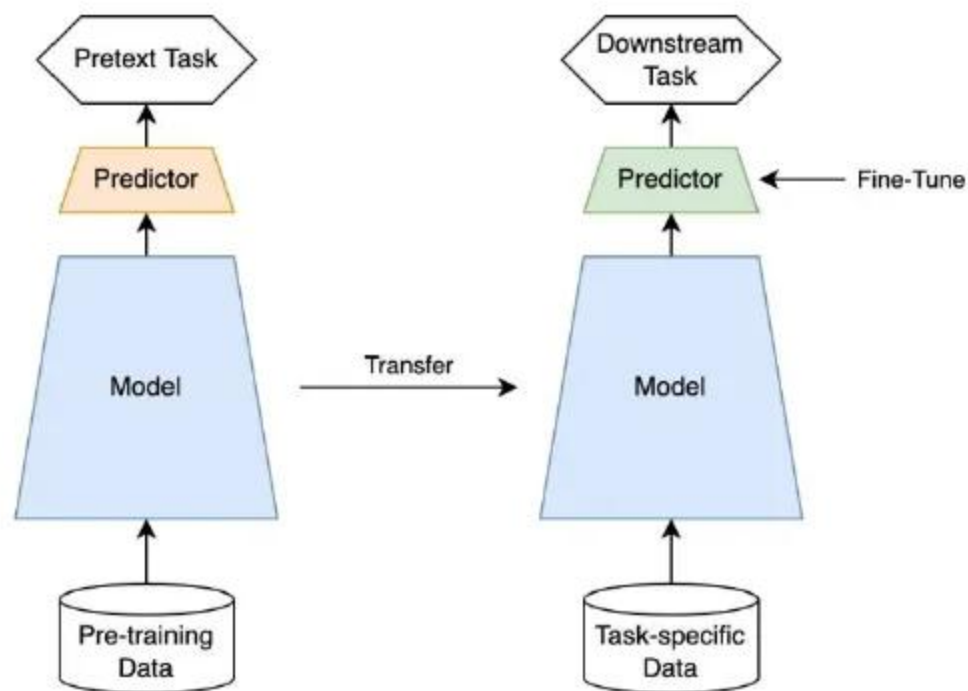
MRR (Mean Reciprocal Rank): Average of the reciprocal ranks of the correct item. The rank is 1 for the first correct item, 1/2 for the second, etc. (Higher is better $\uparrow$, Max 1)

Experiments



Experiments

3. Downstream Tasks : IMU-based Human activity recognition



Models		Ego4D		Aria	
		F1	Acc.	F1	Acc.
Random Init. IMU Encoder		23.23	49.92	56.35	76.11
IMU2CLIP (i ↔ v)	+ Zeroshot	19.39	23.08	18.46	21.52
	+ Probing	40.55	61.46	62.52	83.54
	+ Fine tuning	43.07	65.87	61.77	82.31
IMU2CLIP (i ↔ t)	+ Zeroshot	31.89	36.38	-	-
	+ Probing	45.12	58.01	-	-
	+ Fine tuning	45.15	63.14	-	-
IMU2CLIP (i ↔ v ↔ t)	+ Zeroshot	27.24	24.26	-	-
	+ Probing	42.16	55.21	-	-
	+ Fine tuning	44.17	62.66	-	-

Future Work

Exploring Multimodal Reasoning with LLMs

- Leverage **IMU2CLIP's sensor-to-text translation capability** to enable complex multimodal reasoning tasks.
- **Integrate IMU-derived textual descriptions with Large Language Models (LLMs)** to answer summarization and memory recall queries about user activities.
- This approach offers a promising direction for developing low-power, video-free contextual AI in wearable devices.

```
[15:44] Motion: no activity, Audio: no activity  
[15:45] Motion: walking, Audio: TV noise  
[15:50] Motion: sits down, Audio: watching TV  
[15:52] Motion: look around, Audio: kitchen sounds  
[16:03] Motion: walking, Audio: no activity
```

Question: how long did I sit down and focus on watching TV?

Answer: From 15:50 to 16:03, for a total of 13 minutes.

Why?

IMU2CLIP: IMU → Text



Contribution

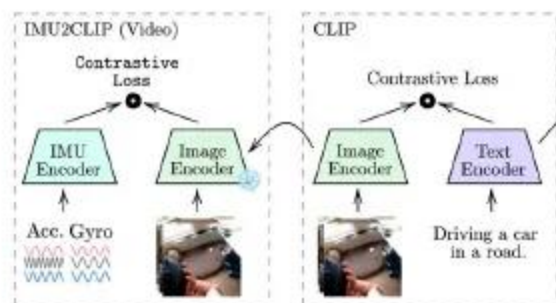
Key Contributions of this paper

1. A novel **large-scale pre-training approach for IMU sensors is proposed**.
2. **An in-depth empirical analysis** evaluating the pre-trained models is provided for both upstream and downstream fine-tuning tasks.
3. Lastly, **novel applications are presented** that show the feasibility of a wider usage for IMU sensor signals.

Limitation & Points to Consider

1. Limitations

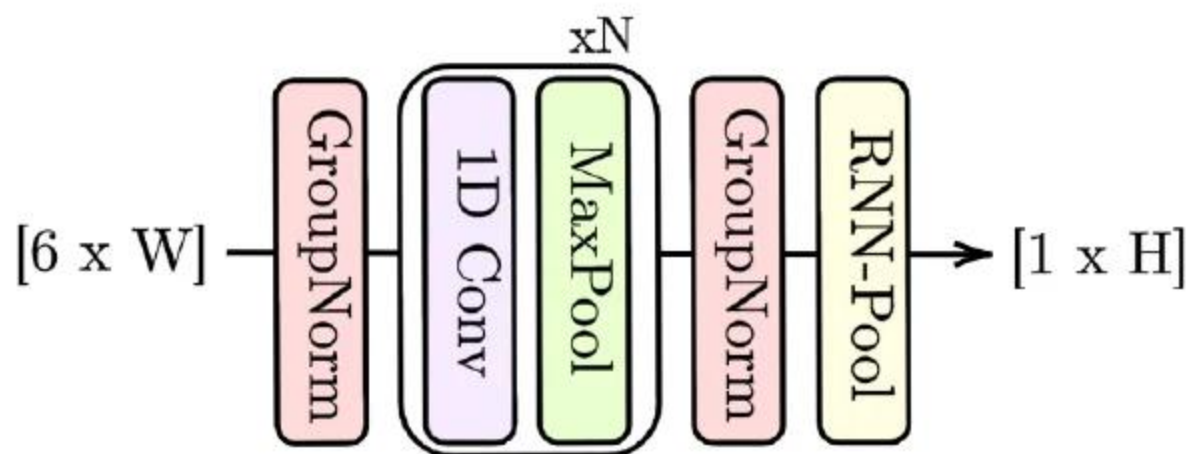
- **Frozen CLIP Encoders:** The reliance on pre-trained and frozen CLIP encoders for video-text alignment might limit the model's ability to fully adapt to or learn novel inter-modal relationships specific to IMU data. This could potentially restrict the nuanced learning of video-text context from the perspective of motion.



2. Points to Consider

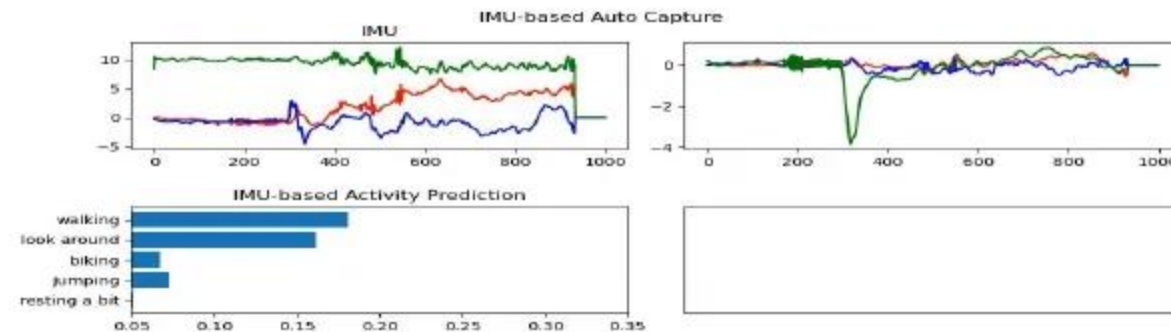
- **Generalization to Third-Person Views:** The current evaluation primarily uses egocentric (first-person) datasets. Future work should investigate the model's performance and generalizability when applied to third-person video data, which represents different visual perspectives and motion patterns.

Appendix



IMU $\rightarrow$ Text Retrieval		
IMU Architecture	MRR	R@50
MW2: Stacked RNN (main)	0.123	60.42
CNN + Attention + RNN	0.112	57.18
Patch Transformer	0.109	54.15
Patch Bi-RNN	0.112	56.21
Patch RNN	0.104	50.83

Appendix



IMU-Predicted Activity: walking

